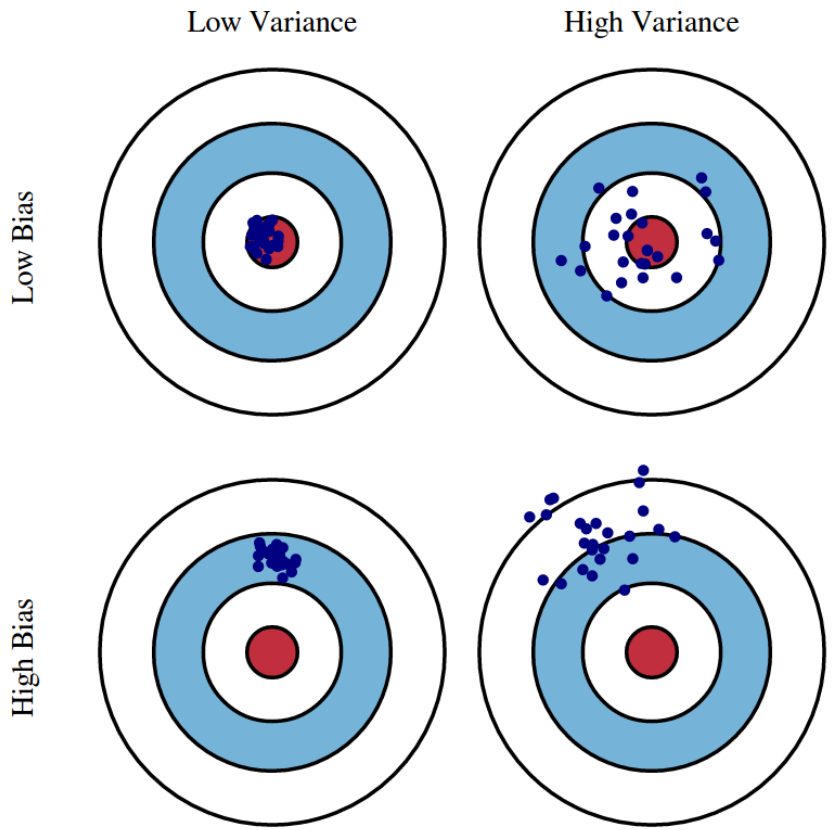
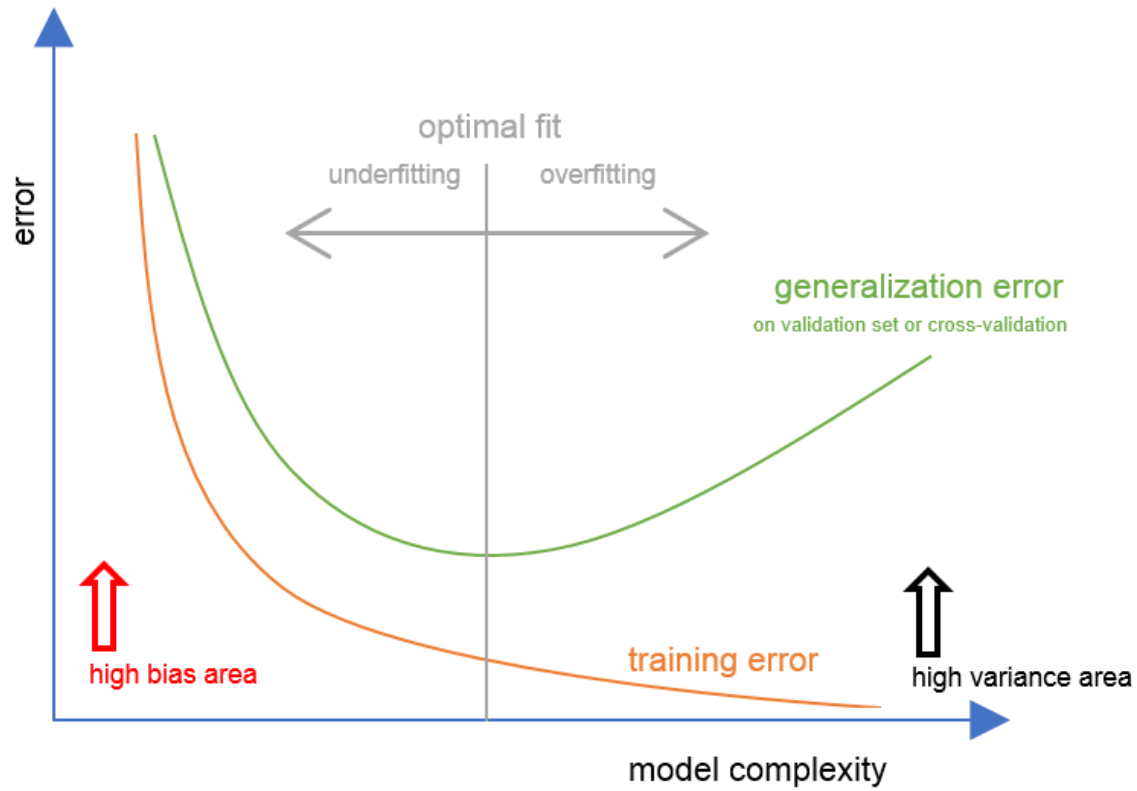
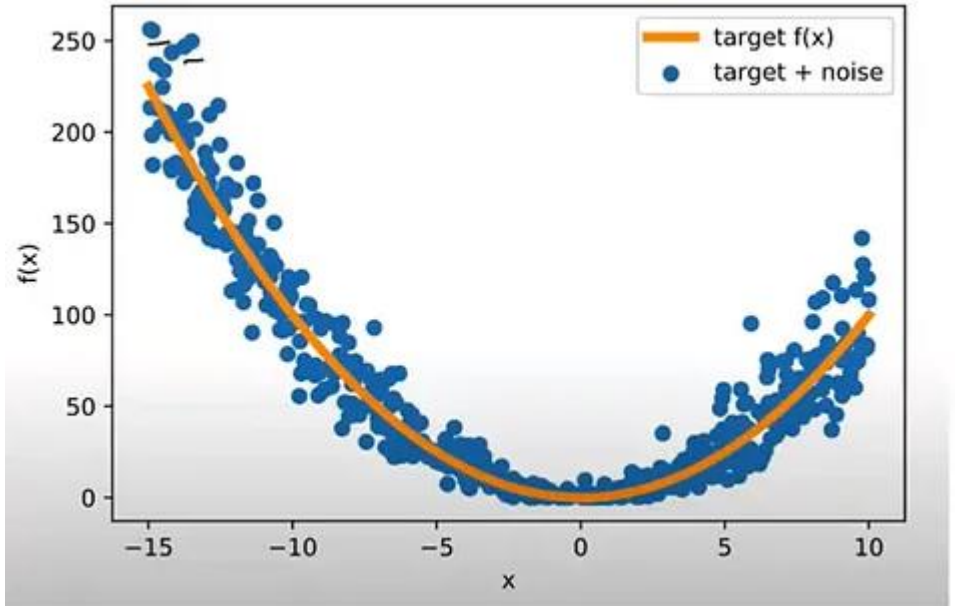
Bias Variance Tradeoff

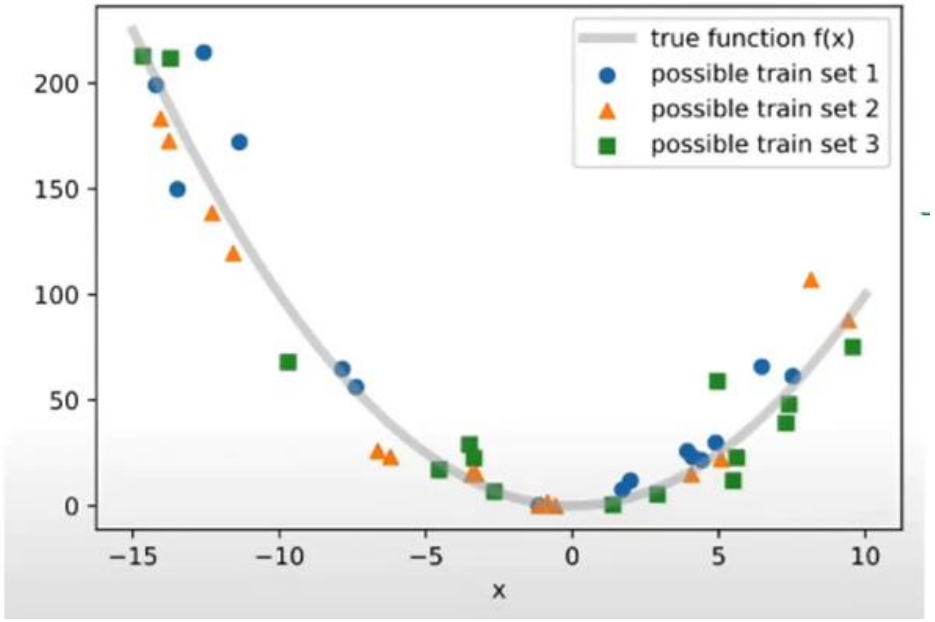
Bias is the difference between the average prediction of our model and the correct value which we are trying to predict.

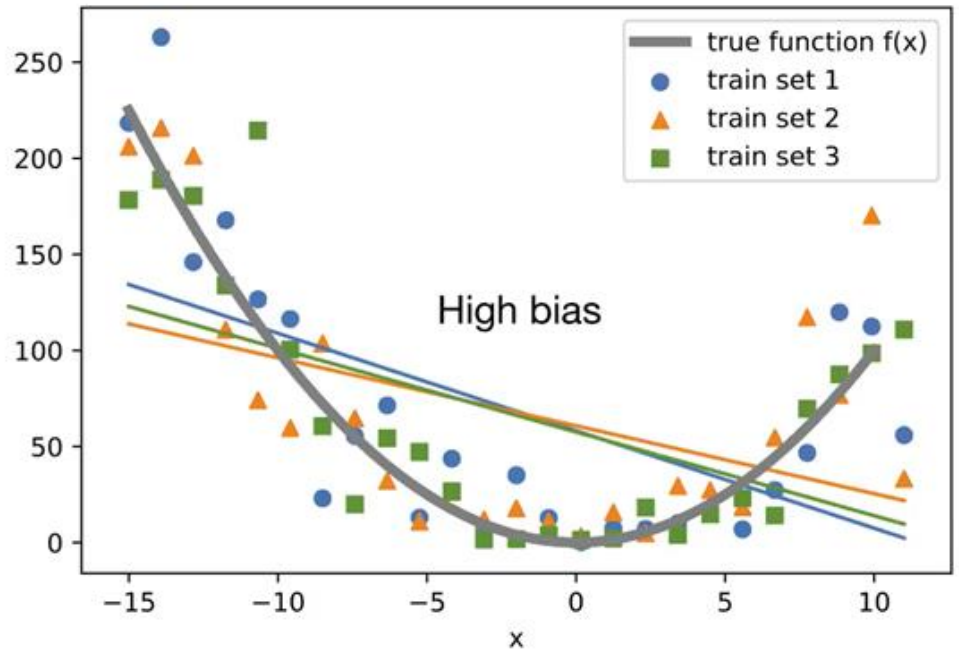
Variance is the variability of model prediction for a given data point or a value which tells us spread of our data.

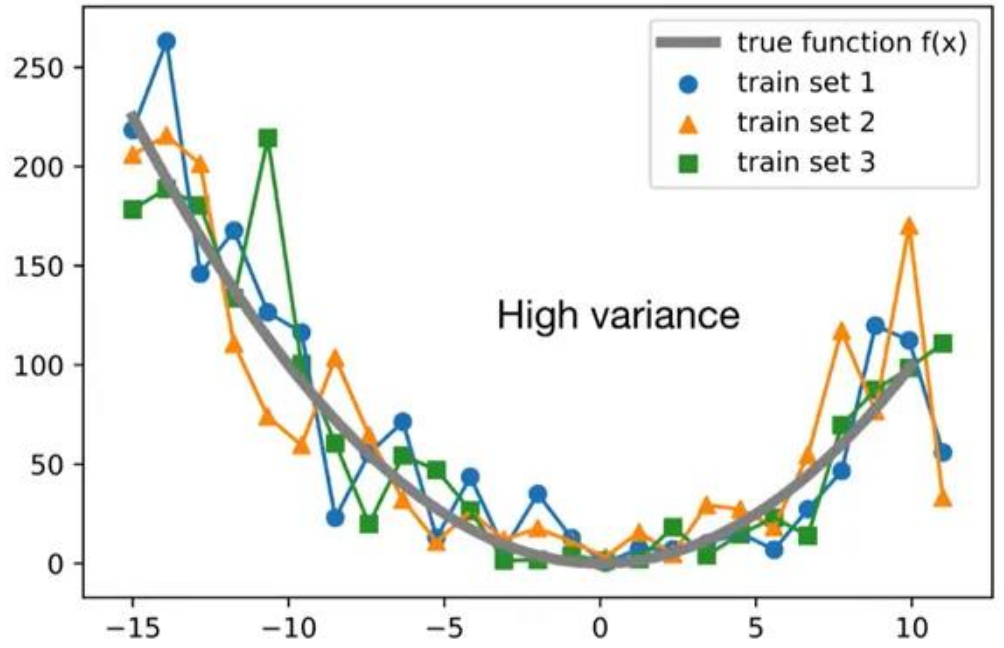








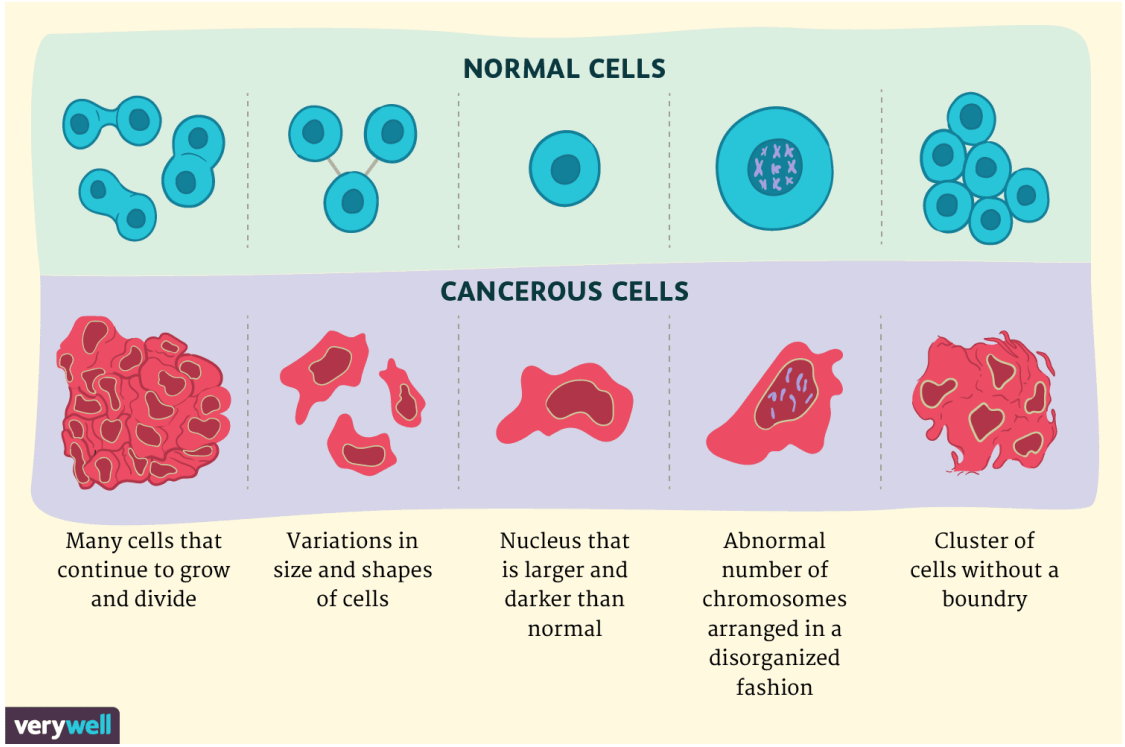




https://medium.com/@sachinsoni600517/mystery-behind-overfitting-and-underfitting-in-machine-learning-bias-variance-trade-off-71bc96b2e8b6

Evaluation Matrices

Accuracy is an evaluation metric that allows you to measure the total number of predictions a model gets right.



Normal cells: 90, Cancerous cells :10

Normal cells correctly identified : 90

Cancerous cells correctly identified : 0

Accuracy of the classifier : 90%

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions}$$

Accuracy is a good metric when classes are balanced and the cost of errors is similar across classes.

When Accuracy Might Be Misleading: **Imbalanced Datasets, Misclassification Costs, Multiclass and Multilabel Problems.**

A confusion matrix is a table used to evaluate classification models, showing how well a model predicted actual outcomes by breaking down correct and incorrect classifications (True Positives, True Negatives, False Positives, False Negatives)

		Predicted	
		Negative	**Positive**
Actual	**Negative**	True Negative	False Positive
	Positive	False Negative	True Positive

Using a confusion matrix we can get the values needed to compute the accuracy, precision and recall of a model.

$$Accuracy = \frac{TrueNegatives + TruePositive}{TruePositive + FalsePositive + TrueNegative + FalseNegative}$$

https://medium.com/@erika.dauria/accuracy-recall-precision-80a5b6cbd28d

Precision and **Recall** are two key metrics derived from the confusion matrix, but they measure different aspects of a model's performance:

Precision measures **how many of the *predicted positives* were actually correct**, focusing on the accuracy of positive predictions.

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive}$$

High Precision: Few false positives.

Use Case: When **false positives** are costly or dangerous, precision becomes more important.

Spam Email Detection: Sending a legitimate email to the spam folder can frustrate users.

Recall (Sensitivity or True Positive Rate)

Definition: Recall measures **how many of the *actual positives* were successfully found**, focusing on capturing all relevant items

$$Recall = \frac{TruePositive}{TruePositive + FalseNegative}$$

High Recall: Few false negatives.

Use Case: When **false negatives** are costly or dangerous, recall becomes more important.

Medical Diagnosis: Missing a disease (false negative) could have life-threatening consequences.

$$F1 = 2 \times \frac{Precision * Recall}{Precision + Recall}$$

When you need to balance precision and recall, especially if the cost of false positives and false negatives is roughly equal, you can use the F1 score

$$\text{Precision} = \frac{True\ Positive}{True\ Positive + False\ Positive}$$

$$\text{Recall} = \frac{True\ Positive}{True\ Positive + False\ Negative}$$

Predicted / Classified

		Normal	**Cancerous**
Actual	Normal	30	20
	Cancerous	30	20

$$Precision = \frac{20}{20+20} = 0.5$$

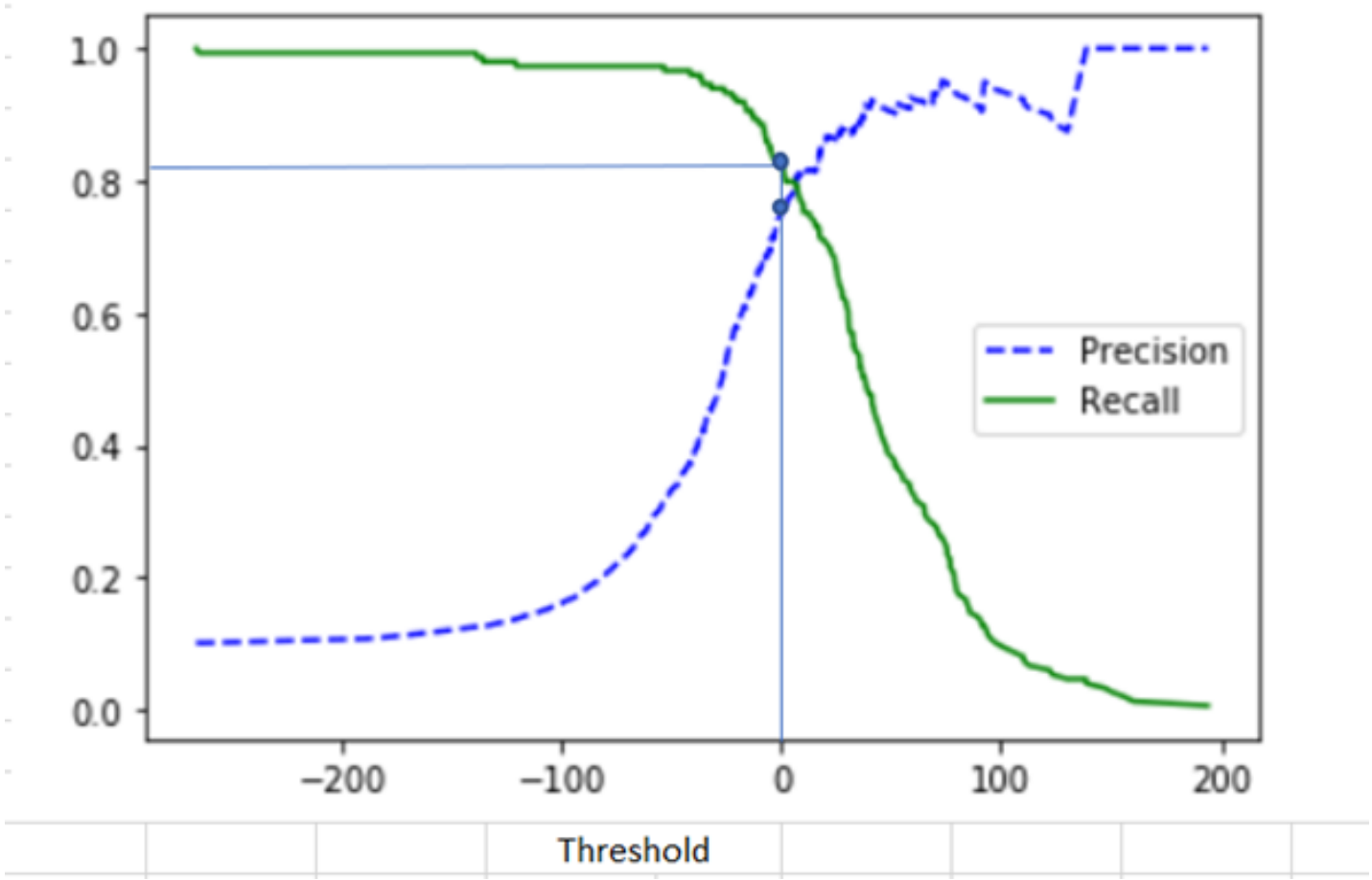
$$Recall = \frac{20}{20+30} = 0.4$$

Predicted

Actual		**Negative**	**Positive**
	Negative	True Negative	False Positive
	Positive	False Negative	True Positive

Precision/Recall Tradeoff

Unfortunately, when we try to improve precision, recall may come down, and vice versa. This is called the *precision/recall tradeoff*.



https://medium.com/analytics-vidhya/precision-recall-tradeoff-79e892d43134

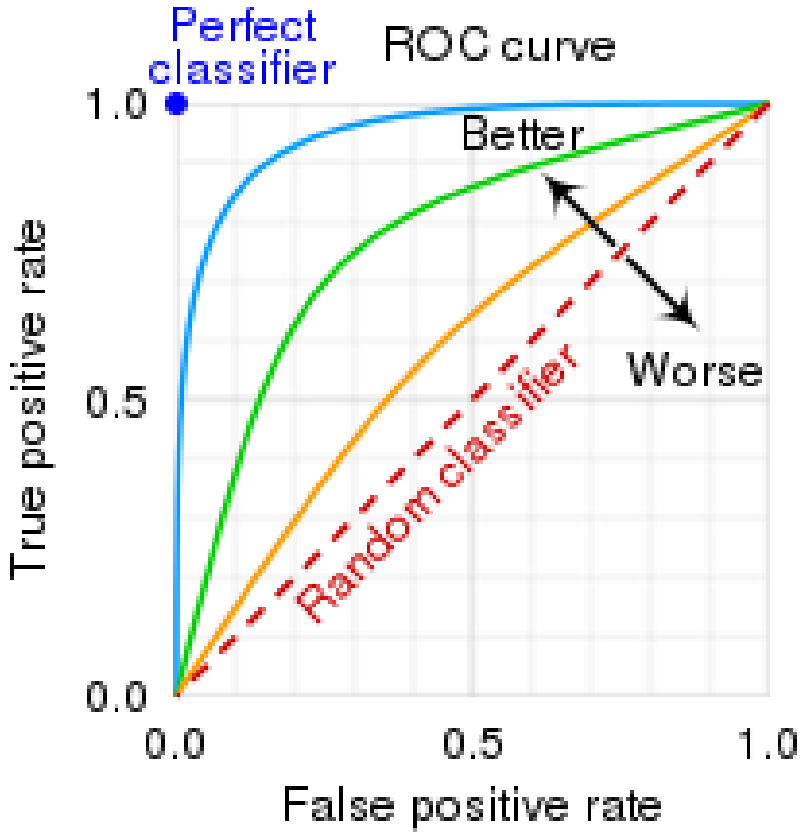
ROC curve

A **receiver operating characteristic curve**, or **ROC curve**, is a graphical plot that illustrates the diagnostic ability of a binary classifier system as its discrimination threshold is varied.

$$\text{TPR /Recall / Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

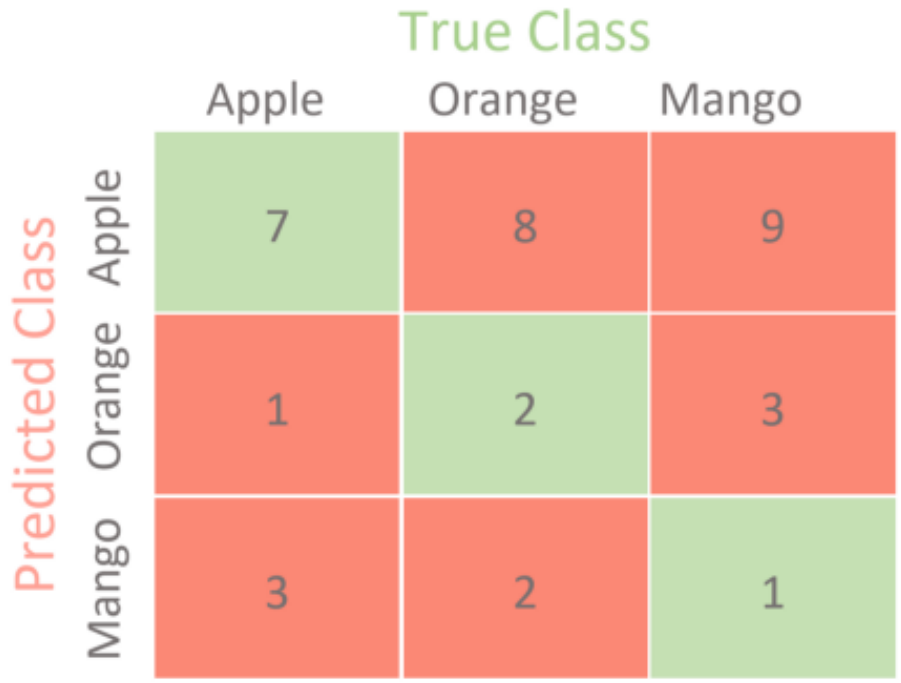
$$\text{FPR} = 1 - \text{Specificity} = \frac{FP}{TN + FP}$$



AUC stands for "Area under the ROC Curve." That is, AUC measures the entire two-dimensional area underneath the entire ROC curve from (0,0) to (1,1).

AUC ranges in value from 0 to 1. A model whose predictions are 100% wrong has an AUC of 0.0; one whose predictions are 100% correct has an AUC of 1.0.

Confusion Matrix for Multi-Class Classification



Class	Precision	Recall	F1-score
Apple	0.29	0.64	0.40
Orange	0.33	0.17	0.22
Mango	0.17	0.08	0.11

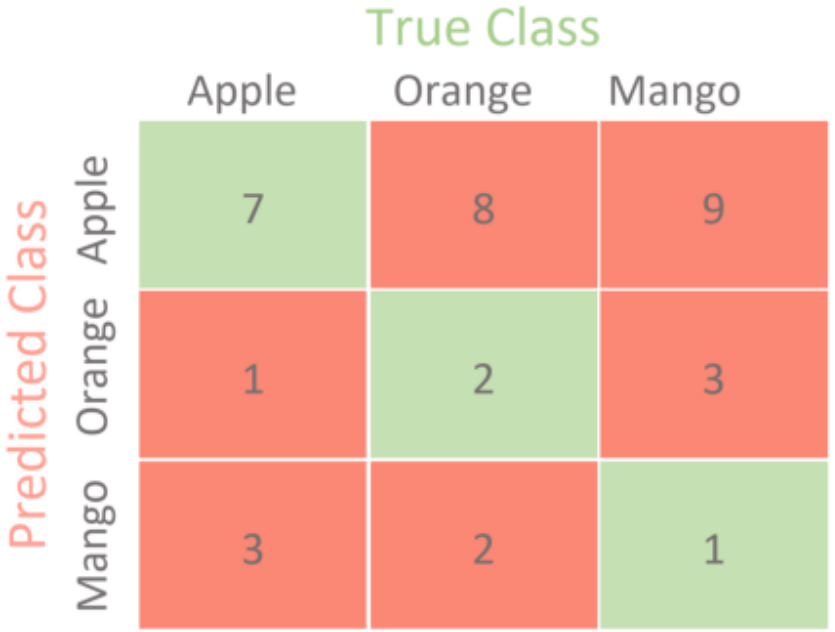
- ***TP = 7***
- ***TN = (2+3+2+1) = 8***
- ***FP = (8+9) = 17***
- ***FN = (1+3) = 4***

- ***Precision = 7/(7+17) = 0.29***
- ***Recall = 7/(7+4) = 0.64***
- ***F1-score = 0.40***

Unlike binary classification, there are no positive or negative classes here.

What we have to do here is to find TP, TN, FP and FN for each individual class.

Source : https://towardsdatascience.com/confusion-matrix-for-your-multi-class-machine-learning-model-ff9aa3bf7826



We can also combine the F1-score of each class to have a single measure for the whole model.

Class	Precision	Recall	F1-score
Apple	0.29	0.64	0.40
Orange	0.33	0.17	0.22
Mango	0.17	0.08	0.11

Micro F1

This is called micro-averaged F1-score. It is calculated by considering the total TP, total FP and total FN of the model.

- *Total TP = (7+2+1) = 10*
- *Total FP = (8+9)+(1+3)+(3+2) = 26*
- *Total FN = (1+3)+(8+2)+(9+3) = 26*

- ***Precision = 10/(10+26) = 0.28***
- ***Recall = 10/(10+26) = 0.28***

Micro F1 = 0.28

Macro F1

- *Class Apple F1-score = 0.40*
- *Class Orange F1-score = 0.22*
- *Class Mango F1-score = 0.11*

Macro F1 = (0.40+0.22+0.11)/3 = 0.24

Weighted F1

It takes a weighted mean of the measures.

Weighted F1 = ((0.40*11)+(0.22*12)+(0.11*13))/(11+12+13) = 0.24